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# Finite-Support Convolutional Likelihoods for Aggregate-Supervised Multiple-Instance Learning

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Anonymous Author(s)

Affiliation

Address

email

## Abstract

Aggregate-supervised multiple-instance learning observes a label for each bag but not the labels of individual instances. Exact likelihoods are known for standard Bernoulli count labels, but many aggregate labels involve more general finite supports, including signed counts with cancellation and integer-valued digit sums. We introduce FS-Conv, a backbone-agnostic finite-support convolutional likelihood layer that maps per-instance contribution PMFs to the exact model-implied aggregate distribution by discrete convolution. FS-Conv strictly generalizes the exact Bernoulli count likelihood underlying recent count-based weak-supervision methods: the  $\{0, 1\}$  support recovers ordinary count loss, while signed and integer-valued supports yield exact likelihoods for cancellation and digit-sum supervision. Across aggregate-supervised prediction tasks, FS-Conv improves hidden-instance recovery most clearly when the observed aggregate is poorly captured by a mean or proportion alone.

## 1 Introduction

Multiple-instance learning (MIL) studies settings where supervision is observed for a bag while the labels of individual instances remain latent. Classical binary MIL uses a positive bag label to indicate that at least one instance is positive. Many aggregate-supervision settings provide more structured labels: the number of positive instances, a class histogram, a signed count, or a scalar integer-valued sum. Similar aggregate labels arise in algebraic combinatorics and mathematical physics, where latent contributions often have combinatorial or algebraic structure (Appendix B). The central problem is to learn an instance-level predictor from such aggregate observations without using instance labels during training.

FS-Conv is a finite-support convolutional likelihood layer for aggregate-supervised MIL. Each instance  $x_i$  is mapped to a probability mass function over a finite set of integer-valued contributions. The layer returns the mathematically implied distribution of the observed aggregate label by convolving the instance-level atomic PMFs. For Bernoulli contributions this recovers the standard count probability used in prior count-loss work. The same construction also gives exact likelihoods for signed counts, digit-sum labels, and one-vs-rest class-count objectives. We refer to this family as finite-support aggregate likelihoods: the atomic support size and contribution values define the aggregate-supervision task, while FS-Conv is unchanged. The formulation separates the instance model from the aggregate likelihood: a backbone model outputs instance-level scores, and the likelihood layer converts these scores into finite-support atomic PMFs before computing the aggregate likelihood. The method is therefore not tied to a particular architecture. Changing the aggregate label changes only the contribution map and aggregate support; the instance backbone, optimizer, and aggregation rule remain unchanged.

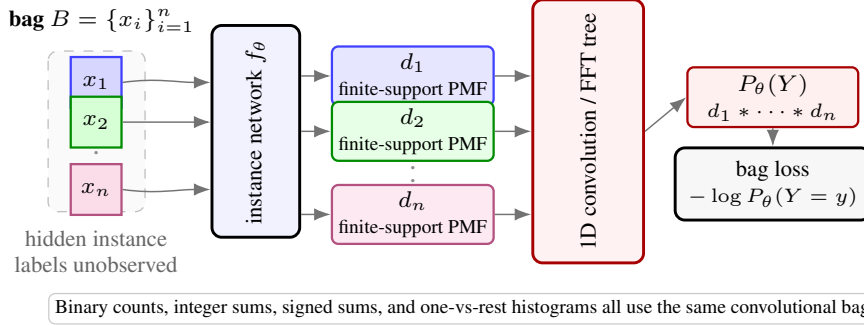


Figure 1: Convolutional aggregate pipeline. The instance network emits finite-support atomic PMFs  $d_i$  for each instance. A deterministic convolutional layer computes the aggregate PMF  $P_\theta(Y)$ , and the observed bag label selects the likelihood term used for training.

This view also makes the novelty boundary explicit. Binary Bernoulli counts recover existing count-loss objectives, and one-vs-rest class-count likelihoods are closely related to the count-likelihood component of LLP-PVC. Our focus is the finite-support likelihood layer that covers these cases while also extending to signed and integer-valued sum labels. FS-Conv is thus a finite-support generalization of the exact count-likelihood component used in recent state-of-the-art count-based weak-supervision and LLP methods:  $\{0, 1\}$  atoms recover ordinary count loss, classwise one-vs-rest atoms recover the LLP-PVC count-likelihood component, and signed or integer-valued atoms extend the same rule beyond Bernoulli counts.

Our contributions are:

- We show that exact Bernoulli count likelihoods used in recent count-based weak-supervision methods are the  $\{0, 1\}$  instance of a broader finite-support convolutional likelihood family.
- We extend this count-likelihood component to signed counts, digit-sum labels, and one-vs-rest class-count objectives by changing only the finite-support contribution map.
- We implement practical convolutional backends, including grouped signed convolution and a GPU FFT-tree backend that parallelizes large multiclass one-vs-rest PMFs beyond CPU dynamic programming.
- We evaluate hidden-instance recovery across Bernoulli count, signed-count, digit-sum, SVHN sum, and one-vs-rest class-count settings.

Empirically, the largest improvements occur for aggregate types where expectation matching discards information, especially digit sums and signed counts. The CIFAR-10 comparison shows that one-vs-rest count likelihood achieves high instance accuracy when paired with ImageNet-pretrained features, but the fixed-instance-budget result attributes the large-bag improvement to greater sampled image exposure rather than to bag size alone.

## 2 Related work

**Multiple-instance learning and count loss.** Classical multiple-instance learning supervises bags rather than instances [8, 19, 1, 6]; neural and attention MIL learn bag predictors end-to-end [28, 11]. These MIL models usually target binary bag-presence labels, such as whether at least one positive instance is present, whereas our aggregate labels are exact integer counts, signed sums, or integer-valued sums. Count-based aggregate supervision gives a stronger label by observing the number of positive instances. Prior count-loss methods compute exact Bernoulli count probabilities and have achieved state-of-the-art or highly competitive results in count-based weak supervision [23]; related count supervision appears in temporal localization [21]. FS-Conv contains these objectives as the  $\{0, 1\}$  special case and lifts the same exact-likelihood mechanism to signed cancellation and integer-valued sums.

**Learning from label proportions.** Learning from label proportions observes class proportions or histograms over bags [20, 31], with probabilistic, deep, pseudo-labeling, consistency, and two-stage variants [25, 3, 10, 9, 14, 26, 15, 30, 16] and recent theoretical analyses [22, 32, 4, 29, 2, 5]. LLP-PVC [17] reframes label proportions as proportional-value classification and uses efficient count-likelihood computation. Our classwise count objective recovers this likelihood component, not the entire LLP-PVC method or a new full joint histogram model.

**Efficient aggregate inference.** Dynamic programming computes Bernoulli count probabilities exactly but is sequential and can be expensive for large bags and many classes [17]. The same finite-support likelihood can instead be evaluated by direct convolution or FFT polynomial multiplication [7]. We use sequential convolution for moderate aggregate supports and a GPU FFT-tree backend for large one-vs-rest count PMFs; this changes the backend, not the statistical objective.

### 3 Problem setup

We study aggregate-supervised bags of instances. A training example is a bag  $B = (x_1, \dots, x_n)$  together with an observed aggregate label  $y$ . The instance-level variables that explain  $y$  are hidden during training and are used only for evaluation. The aggregate label may be a Bernoulli count, a signed count, an integer-valued sum, or a class histogram. In all cases, each instance contributes a finite-support discrete value, and the observed label is the sum of these contributions. We define FS-Conv as one likelihood layer for this structure, so different aggregate-supervision tasks require only different atomic supports and contribution maps.

The unifying object is the *atomic support*. For a one-dimensional task, choose a finite set

$$\mathcal{A} = \{v_0, \dots, v_{S-1}\} \subset \mathbb{Z}, \quad (1)$$

where  $S = |\mathcal{A}|$  is the atomic support size. A backbone model produces scores for each instance, and a contribution map turns those scores into an atomic PMF over  $\mathcal{A}$ . Changing  $\mathcal{A}$  changes the aggregate-supervision task while leaving the aggregate-prediction algorithm unchanged:  $S = 2$  gives Bernoulli counts,  $S = 10$  gives digit sums over  $\{0, \dots, 9\}$ , and  $S = 3$  gives signed atoms over  $\{-1, 0, +1\}$ . For bag size  $n$  and contiguous support  $\{0, \dots, S-1\}$ , the aggregate PMF has length

$$T = n(S-1) + 1. \quad (2)$$

Signed supports use the same principle with an offset, and more general atoms can use instance-dependent supports.

Let  $C_i$  denote the latent contribution of instance  $x_i$ . The contribution may also depend on observed side information  $a_i$ , such as a known sign in signed-count experiments. We assume

$$C_i \in \mathcal{S}_i \subset \mathbb{Z}^d, \quad |\mathcal{S}_i| < \infty, \quad (3)$$

and the observed aggregate is

$$Y = \sum_{i=1}^n C_i. \quad (4)$$

Most experiments use  $d = 1$ . For histograms,  $d$  may be the number of classes, although our scalable implementation uses one-dimensional one-vs-rest count likelihoods. The learner observes  $(x_{1:n}, a_{1:n}, y)$  when side information is present, but never observes the individual  $C_i$  values during training. When discussing individual scalar tasks, we write  $Y_{\text{count}}$  for ordinary nonnegative counts,  $Y_{\text{sum}}$  for integer-valued sums such as digit sums, and  $Y_{\pm}$  for signed counts. These are task-specific instances of the generic aggregate  $Y$  in Eq. (4).

**Definition 1** (FS-Conv finite-support convolutional likelihood layer). *Let  $f_{\theta}$  be any instance model, and let  $\Phi$  be a contribution map that converts the output of  $f_{\theta}$  on instance  $x_i$  and observed side information  $a_i$  into an atomic PMF*

$$d_i = \Phi(f_{\theta}(x_i), a_i) \in \Delta(\mathcal{S}_i), \quad \mathcal{S}_i \subset \mathbb{Z}^d, \quad |\mathcal{S}_i| < \infty. \quad (5)$$

*For conditionally independent latent contributions  $C_i \sim d_i$ , FS-Conv is the deterministic map*

$$\text{Agg}(d_1, \dots, d_n)(y) = (d_1 * \dots * d_n)(y) = \sum_{\substack{c_1 \in \mathcal{S}_1, \dots, c_n \in \mathcal{S}_n \\ c_1 + \dots + c_n = y}} \prod_{i=1}^n d_i(c_i), \quad (6)$$

*which assigns the exact probability mass of the aggregate  $Y = \sum_i C_i$  to value  $y$ .*

112 For Bernoulli atoms  $d_i = [1 - p_i, p_i]$  on  $\{0, 1\}$ , FS-Conv recovers the count-loss probability

$$\Pr_{\theta}(Y = k) = (d_1 * \dots * d_n)(k) = [t^k] \prod_{i=1}^n ((1 - p_i) + p_i t), \quad (7)$$

113 the Poisson-binomial probability optimized by Bernoulli count-loss methods. Under identical in-  
 114 stance probabilities, FS-Conv and CountLoss define the same loss and gradients up to backend dif-  
 115 ferences. Applying the same equation independently to each class with atoms  $[1 - p_{ic}, p_{ic}]$  recovers  
 116 the one-vs-rest count-likelihood component used in LLP-PVC-style objectives, but not a normalized  
 117 full joint multiclass histogram likelihood.

118 Definition 1 separates four choices often coupled in task-specific aggregate-supervision losses:

- 119 • *Backbone independence.* The network supplies atomic PMFs; the likelihood layer is un-  
 120 changed across MLPs, CNNs, transformers, or pretrained encoders.
- 121 • *Contribution-map independence.* Different aggregate-supervision tasks are specified by the  
 122 contribution map and finite support, not by changing the aggregation rule.
- 123 • *Backend independence.* The returned PMF supports NLL training, while its computation  
 124 can be implemented by dynamic programming, direct convolution, grouped convolution,  
 125 or FFT-tree multiplication.
- 126 • *Conditional-independence scope.* The PMF is exact under conditionally independent latent  
 127 contributions; otherwise it is a structured aggregate-likelihood approximation.

128 We train with the aggregate negative log likelihood  $-\log \text{Agg}(d_1, \dots, d_n)(y)$  for the observed label  
 129  $y$ .

130 Once the atomic contribution distributions  $d_i$  are predicted, the aggregate-label distribution is fixed  
 131 by convolution under the conditional-independence model. Tasks differ in the support  $S_i$ , the contri-  
 132 bution map  $\Phi$ , and which side information is observed; concrete kernels are summarized in Table 1.  
 133 Bernoulli count supervision is the simplest case of Eq. (4), while signed counts, integer-valued sums,  
 134 multiplicity counts, and one-vs-rest class counts change the finite-support kernel but keep the same  
 135 likelihood layer. In our experiments, bags are sampled on the fly with replacement from the corre-  
 136 sponding train or test split. Thus an image may appear in multiple bags, but train and test bags are  
 137 always drawn from disjoint dataset splits.

## 138 4 Convolutional bag prediction

### 139 4.1 Aggregate likelihood

140 The instance network and contribution map emit the atomic PMFs  $d_i$  used in Definition 1. For  
 141 one-dimensional contributions, this is a categorical distribution over a finite integer support:

$$d_i(k) = p_{\theta}(z_i = k \mid x_i), \quad k \in \{a, \dots, b\}. \quad (8)$$

142 The exact aggregate-label likelihood is FS-Conv applied to these atomic PMFs:

$$p_{\theta}(Y = y \mid x_{1:n}) = \text{Agg}(d_1, \dots, d_n)(y) = (d_1 * d_2 * \dots * d_n)(y). \quad (9)$$

143 Training minimizes the negative log likelihood

$$\mathcal{L}_{\text{agg}}(\theta) = -\log p_{\theta}(Y = y \mid x_{1:n}). \quad (10)$$

144 The support of the aggregate PMF is finite and contiguous:

$$\text{supp}(Y) = \{na, na + 1, \dots, nb\}. \quad (11)$$

145 Equivalently, if each atom has contiguous support size  $S = b - a + 1$ , then the aggregate PMF has  
 146 width  $T = n(S - 1) + 1$  after shifting by  $na$ . For scalar aggregate labels, all experiments evaluate the  
 147 exact model-implied likelihood over this support; approximation enters through the neural instance  
 148 model rather than through sampled latent assignments.

Table 1: Task kernels used by FS-Conv. The support order is the array order for the atomic PMF.

| Task                      | Aggregate label               | Atomic kernel                                                            | Support               |
|---------------------------|-------------------------------|--------------------------------------------------------------------------|-----------------------|
| Binary count              | $Y = \sum_i z_i$              | $[1 - p_i, p_i]$                                                         | $\{0, 1\}$            |
| Signed count              | $Y_{\pm} = \sum_i s_i z_i$    | $\kappa_i^{(+)} = [0, 1 - p_i, p_i]; \kappa_i^{(-)} = [p_i, 1 - p_i, 0]$ | $\{-1, 0, +1\}$       |
| Integer sum               | $Y_{\text{sum}} = \sum_i z_i$ | $[p_{i0}, \dots, p_{i,K-1}]$                                             | $\{0, \dots, K - 1\}$ |
| Classwise composite count | $h_c = \sum_i z_{ic}$         | $[1 - p_{ic}, p_{ic}]$ for each class $c$                                | $\{0, 1\}$ per class  |

**Two immediate consequences.** First, when the atomic support is Bernoulli, the convolutional likelihood is exactly the coefficient of the polynomial  $\prod_i ((1 - p_i) + p_i t)$  and therefore recovers the standard count probability. Second, for any finite support, the likelihood uses the entire induced aggregate distribution, not only the mean aggregate value. Thus two instance predictors with the same expected sum can receive different likelihoods if they assign different probability mass to the observed aggregate. This is the distinction exploited by the digit-sum and signed-count experiments.

## 4.2 Training objective

For each minibatch, the instance model and task-specific contribution map produce atomic PMFs for every instance in every bag. FS-Conv then maps these atoms deterministically to an exact PMF over the observed aggregate label. This separation makes the objective backbone-agnostic: an MLP, CNN, transformer, or pretrained image encoder can be used as long as it produces per-instance scores from which  $\Phi$  can form the required finite-support PMFs. It also makes the aggregate-supervision family modular: Bernoulli counts, signed counts, integer-valued sums, and one-vs-rest class counts use different kernels but the same likelihood rule. The training loss is the negative log probability assigned to the observed aggregate,

$$\mathcal{L}(\theta) = -\frac{1}{B} \sum_{b=1}^B \log (d_{b1} * \dots * d_{bn_b}) (y_b). \quad (12)$$

Gradients are backpropagated through FS-Conv into the instance network. Instance labels are never used for training in the aggregate-supervision experiments; they are used only for evaluation of hidden instance recovery. The expected-value baseline regresses only  $\sum_i \mathbb{E}[z_i]$  to the observed aggregate, so it discards all information beyond the first moment.

Appendix A gives implementation details, including the classwise composite score used for one-vs-rest counts.

## 4.3 Special cases

Table 1 summarizes the task-specific kernels; FS-Conv itself is unchanged.

**Binary counts.** A target concept gives hidden Bernoulli indicators  $z_i$ , and the observed label is  $Y = \sum_i z_i$ . The kernel  $[1 - p_i, p_i]$  recovers the standard Bernoulli count likelihood, so this case is included as a connection to prior count-loss work.

**Signed counts.** With observed signs  $s_i \in \{-1, +1\}$ , target instances contribute  $s_i$  and non-target instances contribute 0, giving  $Y_{\pm} = \sum_i s_i z_i$ . The signed kernels in Table 1 model cancellation explicitly because many latent assignments can yield the same signed sum.

**Integer-valued sums.** For integer-valued labels  $z_i \in \{0, \dots, K - 1\}$ , the atomic kernel is the full categorical vector and the observed label is  $Y_{\text{sum}} = \sum_i z_i$ . The likelihood therefore compares the observed sum to the full convolved PMF rather than only to the expected sum  $\sum_i \mathbb{E}[z_i]$ .

**Histograms.** For multiclass count labels, we use a classwise composite likelihood over one-vs-rest Bernoulli counts: for each class  $c$ , FS-Conv scores the observed count  $h_c = \sum_i z_{ic}$ , and the loss averages over classes. This is scalable, but it is not a full joint multiclass histogram model; it is closely related to the count-likelihood component of LLP-PVC. The corresponding composite score is therefore not a normalized joint histogram probability.

## 186 4.4 Computational backends

187 All of the above objectives are one-dimensional convolutions over integer aggregate support. If  $m^{(i)}$   
188 denotes the aggregate PMF after processing the first  $i$  atoms, then

$$m^{(i)}(s) = \sum_{r=a}^b m^{(i-1)}(s-r)d_i(r), \quad m^{(0)} = \delta_0. \quad (13)$$

189 This recurrence is the same probability law whether implemented as dynamic programming, shifted  
190 dense tensor accumulation, grouped convolution, or FFT polynomial multiplication. The choice  
191 of backend is therefore computational rather than statistical: the likelihood being optimized is un-  
192 changed, but the hardware profile differs. We use sequential convolution for moderate binary, signed,  
193 and integer-valued sum experiments; Shukla-style dynamic programming as a Bernoulli count base-  
194 line; grouped GPU convolution for signed Bernoulli groups; and a balanced FFT-tree backend for  
195 large classwise one-vs-rest count PMFs. Further implementation details and backend complexity  
196 are given in Appendix A.

## 197 5 Experiments

### 198 5.1 Experimental setup

199 We evaluate FS-Conv under five aggregate-supervision themes:

- 200 1. integer-valued sum labels;
- 201 2. signed count labels with cancellation;
- 202 3. Bernoulli count labels as a recovery of prior count loss;
- 203 4. classwise count labels connecting to LLP;
- 204 5. runtime of exact aggregate backends.

205 Unless otherwise stated, MNIST-family experiments use LeNet-style instance networks, bags are  
206 sampled on the fly, and tables report mean  $\pm$  sample standard deviation over random seeds when  
207 multiple seeds are available. Hidden instance labels are never used for training; they are used only  
208 for evaluation. Implementation details are summarized in Appendix A. Each subsection changes the  
209 observed aggregate label while keeping the same finite-support convolutional likelihood layer.

210 **Metrics.** Across the experimental tables, aggregate MAE columns report the mean absolute dif-  
211 ference between the predicted aggregate and the observed aggregate on test bags. For each seed  
212 and configuration, all methods are evaluated on the same deterministic held-out bag sampler; train  
213 and test bags are drawn from disjoint underlying dataset splits. For expected-value baselines, the  
214 predicted aggregate is the scalar expectation trained by the baseline. For FS-Conv rows, the main  
215 aggregate prediction is the mode of the learned aggregate PMF; we also report expected-aggregate  
216 MAE for the main FS-Conv settings where this statistic is directly comparable to expected-value  
217 baselines. For classwise count objectives, count MAE is the mean absolute class-count error aver-  
218 aged over classes and test bags,

$$\frac{1}{|\mathcal{B}_{\text{test}}|} \sum_{B \in \mathcal{B}_{\text{test}}} \frac{1}{C} \sum_{c=1}^C |\hat{h}_{Bc} - h_{Bc}|. \quad (14)$$

219 Rows labeled MSE denote MSE training objectives; MAE and accuracy columns are evaluation  
220 metrics computed after training. Columns labeled instance accuracy use held-out instance labels  
221 only for evaluation, never for training. Instance accuracy is never used for model selection unless  
222 explicitly stated.

### 223 5.2 Integer-valued sum labels

224 Digit-sum supervision is an integer-valued aggregate label: the learner observes a single scalar while  
225 each hidden instance label has ten possible values. Here  $z_i$  is the original MNIST digit value in

Table 2: Integer-valued sum labels. FS-Conv uses the full convolved PMF over sums, whereas MSE regresses only the expected sum. MNIST FS-Conv rows report mean  $\pm$  standard deviation over three seeds; SVHN rows use ResNet-18 backbones and are single-seed diagnostics.

| Data         | $n$ /train bags | FS-Conv MAE      | FS-Conv inst. acc. | MSE MAE | MSE inst. acc. |
|--------------|-----------------|------------------|--------------------|---------|----------------|
| MNIST        | 10/1000         | 0.612 $\pm$ .008 | .982 $\pm$ .000    | 2.535   | .226           |
| MNIST        | 10/5000         | 0.277 $\pm$ .026 | .992 $\pm$ .001    | 1.023   | .746           |
| MNIST        | 50/1000         | 4.901 $\pm$ .973 | .693 $\pm$ .228    | 6.274   | .248           |
| MNIST        | 50/5000         | 2.229 $\pm$ .616 | .887 $\pm$ .102    | 2.594   | .817           |
| SVHN scratch | 10/5000         | 5.812            | .229               | 5.328   | .249           |
| SVHN pretr.  | 10/5000         | 1.324            | .955               | 2.109   | .826           |

Table 3: Signed aggregates with random known signs. MSE regresses the expected signed sum  $\sum_i s_i p_i$ , while FS-Conv uses the full finite-support signed aggregate distribution. MNIST FS-Conv rows report mean  $\pm$  standard deviation over three seeds; FashionMNIST rows report seed 0–1 means; SVHN and CIFAR-10 rows are seed-0 pretrained ResNet-18 experiments. A dash indicates that the corresponding paired MSE baseline was not evaluated.

| Data            | $n$ /train bags | MSE MAE | MSE inst. acc. | FS-Conv MAE      | FS-Conv inst. acc. |
|-----------------|-----------------|---------|----------------|------------------|--------------------|
| MNIST           | 10/1000         | 0.269   | .959           | 0.095 $\pm$ .010 | .990 $\pm$ .001    |
| MNIST           | 10/5000         | 0.047   | .995           | 0.045 $\pm$ .010 | .996 $\pm$ .001    |
| MNIST           | 50/1000         | 0.398   | .991           | 0.243 $\pm$ .053 | .994 $\pm$ .001    |
| MNIST           | 50/5000         | 0.203   | .995           | 0.198 $\pm$ .022 | .996 $\pm$ .000    |
| FashionMNIST    | $n = 10$        | 0.121   | .987           | 0.113            | .988               |
| FashionMNIST    | $n = 50$        | 0.413   | .988           | 0.381            | .990               |
| SVHN pretr.     | $n = 16$        | 0.745   | .926           | 0.684            | .934               |
| SVHN pretr.     | $n = 64$        | 1.639   | .914           | 1.407            | .943               |
| CIFAR-10 pretr. | $n = 16$        | –       | –              | 0.343            | .974               |
| CIFAR-10 pretr. | $n = 64$        | –       | –              | 0.753            | .980               |

226  $\{0, \dots, 9\}$  and the aggregate label is  $Y_{\text{sum}} = \sum_i z_i$ . Exact finite-support likelihood uses the full  
227 induced distribution over possible sums, whereas an expected-value baseline regresses only  $\sum_i \mathbb{E}[z_i]$   
228 to the observed sum.

229 For  $n = 10$  and 1000 training bags, the MSE baseline has much worse aggregate error and hidden  
230 digit recovery than FS-Conv. With more training bags it improves, but FS-Conv remains better in  
231 both aggregate MAE and hidden digit accuracy. For  $n = 50$ , the task is harder for both objectives be-  
232 cause many digit assignments can give similar sums; FS-Conv still has lower aggregate error, while  
233 the lower hidden accuracy shows that large scalar-sum bags are underdetermined. SVHN supports  
234 the same conclusion beyond MNIST when the visual backbone is strong: pretraining makes natural  
235 RGB house-number digit sums usable, and FS-Conv outperforms expected-sum regression with the  
236 same pretrained ResNet-18 backbone. On SVHN with 5000 bags, the pretrained FS-Conv model  
237 reaches 95.46% hidden digit accuracy and 65.00% exact sum accuracy, while the pretrained MSE  
238 baseline reaches 82.58% hidden digit accuracy and 22.58% exact sum accuracy. For the MNIST FS-  
239 Conv rows, expected-aggregate MAE is 0.704 $\pm$ .029, 0.306 $\pm$ .022, 4.864 $\pm$ 1.003, and 2.183 $\pm$ .544  
240 for the four rows in Table 2, respectively.

### 241 5.3 Signed aggregates with cancellation

242 Signed counts test an aggregate that is not reducible to ordinary class proportions. We again use the  
243 digit-9 target indicator, but each instance carries a known random sign; target-digit images contribute  
244 +1 or −1, while all other digits contribute 0. Even with random signs, the observed signed sum is  
245 many-to-one: positive and negative contributions can cancel, so different latent assignments can  
246 yield the same aggregate label. FS-Conv models this full signed aggregate distribution, whereas the  
247 MSE baseline only matches the expected signed sum.

248 Both methods recover accurate target detectors with enough data, but the likelihood uses the full  
249 signed aggregate distribution rather than only  $\sum_i s_i p_i$  and gives lower signed-count error. Fashion-  
250 MNIST shows the same pattern beyond MNIST: MSE is close in instance recovery, but FS-Conv  
251 is consistently better on the signed aggregate. On SVHN with ImageNet-pretrained ResNet-18, the

Table 4: Bernoulli count labels. The Bernoulli count likelihood recovers accurate hidden target detectors on MNIST target-count bags.

| Data  | $n$ /train bags | Method      | Count MAE | Instance acc. |
|-------|-----------------|-------------|-----------|---------------|
| MNIST | 10/1000         | exact count | 0.115     | 0.988         |
| MNIST | 10/5000         | exact count | 0.090     | 0.991         |
| MNIST | 50/1000         | exact count | 0.686     | 0.987         |
| MNIST | 50/5000         | exact count | 0.460     | 0.991         |

Table 5: Main classwise count-supervision results. Count MAE is mean absolute class-count error. FS-Conv denotes the same finite-support convolutional likelihood layer applied as a classwise composite likelihood over one-vs-rest counts, as described in Section 4. A dash indicates that the corresponding paired MSE baseline was not evaluated.

| Data     | $n$ /train bags | Backbone   | MSE MAE          | MSE inst. acc.  | FS-Conv MAE     | FS-Conv inst. acc. |
|----------|-----------------|------------|------------------|-----------------|-----------------|--------------------|
| MNIST    | 50/5000         | LeNet      | 1.684 $\pm$ .000 | .116 $\pm$ .000 | .080 $\pm$ .003 | .992 $\pm$ .000    |
| CIFAR-10 | 16/1000         | pretr. R18 | .702             | .668            | .260            | .902               |
| CIFAR-10 | 64/1000         | pretr. R18 | 1.855            | .272            | .678            | .914               |
| CIFAR-10 | 64/250          | pretr. R18 | —                | —               | 1.115           | .803               |

advantage is modest for  $n = 16$  but clearer for  $n = 64$ , where FS-Conv improves instance accuracy from 91.37% to 94.33% and signed MAE from 1.639 to 1.407. The CIFAR-10 pretrained rows evaluate whether the signed finite-support objective remains usable in a harder natural-image setting; they are not paired with MSE baselines. They reach instance AUCs of 98.23% for  $n = 16$  and 99.19% for  $n = 64$ . This setting illustrates a distinction from nonnegative count supervision: cancellation changes the inverse problem even when the latent instance variable remains Bernoulli. For the MNIST FS-Conv rows, posterior-mean signed MAE is  $0.109 \pm .011$ ,  $0.050 \pm .006$ ,  $0.273 \pm .054$ , and  $0.217 \pm .014$  for the four rows in Table 3, respectively.

#### 5.4 Bernoulli count likelihood as a special case

Bernoulli count supervision validates the connection to prior count-loss work. For MNIST-Bags, we make MNIST binary by choosing digit 9 as the positive class and treating digits 0–8 as negative. The learner observes the number of target-digit images in the bag; hidden instance labels are used only for evaluation. This case verifies that FS-Conv recovers the established Bernoulli count probability while sharing the same implementation pattern as the new aggregate types. Standard attention MIL is not a like-for-like baseline here, because it is typically trained from a binary presence label  $\mathbf{1}\{Y > 0\}$ , whereas this setting observes the full count  $Y$ .

This experiment is included as a recovery check rather than a novelty claim. It confirms that the same likelihood layer used for signed counts and integer-valued sums also gives the standard Bernoulli count likelihood and accurate hidden target recovery.

#### 5.5 Classwise count objectives and LLP connections

We include histogram supervision to position FS-Conv relative to recent learning-from-label-proportions objectives. For MNIST histograms, the hidden category is the original digit class and the observed label is the vector of digit counts in the bag. We evaluate a classwise composite likelihood: for each class, the model scores the observed one-vs-rest count with a Bernoulli count PMF. This corresponds to the count-likelihood component of LLP-PVC, rather than a full joint multiclass histogram model. The comparison here is intentionally limited: MSE matches expected class counts, while the classwise FS-Conv likelihood scores each observed count with the exact one-vs-rest count PMF.

On the MNIST large-bag setting, classwise FS-Conv improves over MSE proportion matching in count MAE and hidden digit accuracy. This supports the connection to class-count LLP objectives without making the histogram setting the main novelty claim. The CIFAR-10 rows are narrower: they show that classwise FS-Conv is effective with ImageNet-pretrained ResNet-18 features and outperforms a pretrained MSE histogram baseline at  $n = 16$  and  $n = 64$ . However, the  $n = 64$ ,



Table 6: Runtime for multiclass one-vs-rest count PMFs with batch size 256 and  $C = 10$  classes. CPU DP is compared to GPU FFT-tree convolution for forward aggregate PMF computation only.

| Bag size | CPU DP (s) | GPU FFT-tree (s) | Speedup           | GPU mem. | Max error            |
|----------|------------|------------------|-------------------|----------|----------------------|
| 64       | 8.246      | 0.00097          | $8.5 \times 10^3$ | 13.8 MB  | $3.4 \times 10^{-7}$ |
| 128      | 17.230     | 0.00097          | $1.8 \times 10^4$ | 28.6 MB  | $4.2 \times 10^{-7}$ |
| 256      | 36.785     | 0.00121          | $3.0 \times 10^4$ | 55.6 MB  | $5.1 \times 10^{-7}$ |
| 512      | 54.040     | 0.00134          | $4.0 \times 10^4$ | 113.3 MB | $6.6 \times 10^{-7}$ |

train 1000 setting exposes four times as many image instances per epoch as  $n = 16$ , train 1000. The fixed-instance-budget setting,  $n = 64$  with 250 train bags, reaches only about 80.3% final instance accuracy. Thus the CIFAR-10 results should be read as evidence that FS-Conv can use aggregate-supervised image volume effectively under a strong representation, not as evidence that larger bags are intrinsically superior.

## 5.6 Backend runtime

Our FFT-tree backend computes batched multiclass one-vs-rest count PMFs on GPU and substantially reduces wall-clock time relative to a CPU dynamic-programming implementation in this benchmark. This addresses a practical limitation of DP-based count losses: exact count-probability computation need not be tied to sequential CPU dynamic programming. The benchmark in Table 6 isolates aggregate PMF evaluation for batch size 256 and  $C = 10$  classes, so each timing evaluates 2560 binary count PMFs. It measures backend speed rather than end-to-end training time or representation quality.

## 6 Discussion

The experiments support FS-Conv as a common likelihood layer for finite-support aggregate supervision. Its strongest gains appear when the aggregate label is poorly summarized by a mean or proportion, as in low-data digit sums and signed counts with cancellation. The Bernoulli and class-wise count experiments primarily establish continuity with prior count-likelihood work: they show that the same interface includes prior count likelihoods and one-vs-rest count pseudo-likelihoods, not that ordinary count loss or full joint histogram LLP is new.

The main limitations are ambiguity and representation dependence. Large scalar-sum bags can admit many latent assignments with the same aggregate value, and signed aggregates make this non-identifiability explicit through cancellation. CIFAR-scale results also depend on pretrained features and sampled image exposure; the fixed-instance-budget comparison shows that larger bags are not intrinsically better.

Future work will use this finite-support signed-count view in the mathematical setting that motivated part of the project. Appendix B sketches how signed aggregate labels arise from conjectural factorizations of  $(q, t)$ -Littlewood–Richardson coefficients. Developing that application requires problem-specific architectures and diagnostics beyond the scope of this paper, but it is a natural next test of whether exact aggregate likelihoods can recover latent combinatorial structure.

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## A Implementation details

All aggregate-supervision experiments sample bags on the fly rather than pregenerating fixed bag files. Within each seed and configuration, paired methods use the same deterministic held-out bag sampler so aggregate MAE and hidden-instance metrics are computed on the same test bags. Instance labels are used only for evaluation metrics such as hidden-instance AUC or hidden digit accuracy. MNIST-family experiments use LeNet-style instance classifiers, Adam with learning rate  $5 \times 10^{-4}$ , weight decay  $10^{-4}$ , and 50 epochs. The main MNIST-family configurations use seeds 0, 1, 2, test on 1000 sampled bags, and evaluate mean bag sizes 10 and 50 with standard deviations 2 and 10.

**Backend implementation.** Sequential finite-support convolution accumulates shifted copies of the current aggregate PMF and is used for the moderate-size binary, signed, and integer-valued sum experiments. Dynamic programming is used as a Shukla-style Bernoulli count baseline and computes the same Bernoulli count coefficients as FS-Conv. For grouped signed binary aggregates, each active group starts as a point mass at zero on support  $[-K_{\max}, K_{\max}]$ . For one signed instance, the grouped GPU backend applies a length-three channelwise kernel and padding one. Because PyTorch’s conv1d operator is cross-correlation rather than mathematical convolution, the implementation stores the kernel in the order

$$[p_i \mathbf{1}\{s_i = +1\}, 1 - p_i, p_i \mathbf{1}\{s_i = -1\}], \quad (15)$$

---

**Algorithm 1** Convolutional aggregate prediction for one minibatch

---

**Require:** Bags  $\{(x_{b1:n_b}, a_{b1:n_b}, y_b)\}_{b=1}^B$ , backbone  $f_\theta$ , contribution map  $\Phi$ , backend Conv  
**Ensure:** Loss  $\mathcal{L}$  and aggregate-label scores  $\{s_b\}_{b=1}^B$

```
1: for  $b = 1, \dots, B$  do
2:    $H_b \leftarrow f_\theta(x_{b1:n_b})$ 
3:   for  $i = 1, \dots, n_b$  do
4:      $d_{bi} \leftarrow \Phi(H_{bi}, a_{bi})$ 
5:   end for
6:   if classwise one-vs-rest count objective then
7:     for  $c = 1, \dots, C$  do
8:        $m_{bc} \leftarrow \text{Conv}(\{[1 - p_{bic}, p_{bic}]\}_{i=1}^{n_b})$ 
9:        $p_{bc} \leftarrow m_{bc}(h_{bc})$ 
10:    end for
11:     $s_b \leftarrow \prod_{c=1}^C p_{bc}^{1/C}$  ▷ composite score
12:     $\ell_b \leftarrow -C^{-1} \sum_{c=1}^C \log p_{bc}$ 
13:  else
14:     $m_b \leftarrow \text{Conv}(\{d_{bi}\}_{i=1}^{n_b})$ 
15:     $s_b \leftarrow m_b(y_b); \ell_b \leftarrow -\log s_b$ 
16:  end if
17: end for
18: return  $\mathcal{L} = B^{-1} \sum_{b=1}^B \ell_b, \{s_b\}_{b=1}^B$ 
```

---

Table 7: FS-Conv backend implementations.  $n$  is bag size,  $S$  is atomic support size,  $T = n(S-1)+1$  is aggregate PMF width, and  $C$  is number of one-vs-rest classes.

| Backend                | Setting                      | Work per bag                | Use                           |
|------------------------|------------------------------|-----------------------------|-------------------------------|
| DP                     | Bernoulli count              | $O(nT)$                     | Shukla-style count baseline   |
| Sequential convolution | finite support               | $O(nST)$                    | main moderate-bag experiments |
| Grouped convolution    | signed Bernoulli             | $O(nT)$ ,<br>batched groups | signed GPU implementation     |
| FFT-tree               | finite support / one-vs-rest | $O(CT \log T \log n)$       | runtime, classwise counts     |

415 so the left kernel entry shifts probability mass toward larger aggregate values and the right entry  
416 shifts mass toward smaller values. This is only an implementation orientation; in mathematical  
417 support order  $[-1, 0, +1]$  the signed kernels are the atoms described in Section 4.

418 For digit-sum, the atom width is the number of finite digit values, for example ten for MNIST  
419 digits. For class-count experiments, we avoid a full  $C$ -dimensional joint histogram over classes  
420 and instead compute  $C$  one-vs-rest one-dimensional count likelihoods in parallel. This keeps the  
421 aggregate support one-dimensional for each class and makes large bags practical on GPU. The  
422 FFT-tree backend is useful for this setting because the same pairwise polynomial products are  
423 batched over classes and bags. Padding or masked positions are represented by deterministic  
424 zero-contribution atoms, such as  $[1, 0]$  for Bernoulli counts,  $[0, 1, 0]$  for signed support  $[-1, 0, +1]$ ,  
425 and  $[1, 0, \dots, 0]$  for integer-valued sums. The main multiclass one-vs-rest runtime benchmark  
426 is produced by `scripts/bench_multiclass_ovr_fft.py`; the reported CUDA run is stored  
427 in `results/bench_multiclass_ovr_fft_cuda_20260505/`. This benchmark compares CPU  
428 Bernoulli count DP with batched GPU FFT-tree convolution for forward aggregate PMF computa-  
429 tion only, not full training.

430 These forward+backward measurements come from `scripts/bench_exactness_runtime.py`.  
431 They show that GPU DP and convolution are close for simple binary counts, while the largest run-  
432 time separation appears when many one-vs-rest PMFs are batched and evaluated by the FFT-tree  
433 backend.

434 **Class-count and CIFAR settings.** Class-count experiments treat each class count as a one-vs-rest  
435 Bernoulli count likelihood and optimize the resulting classwise composite likelihood. The MNIST  
436 class-count experiments use batch size 192 for large-bag configurations. CIFAR-10 experiments use

Table 8: Representative forward+backward GPU runtime for exact aggregate likelihood backends.

| Task                       | Bag size | Time (s) |
|----------------------------|----------|----------|
| Binary DP                  | 32       | 0.0132   |
| Binary convolution         | 32       | 0.0108   |
| Binary DP                  | 256      | 0.0963   |
| Binary convolution         | 256      | 0.0749   |
| Ordinal-10 sum convolution | 32       | 0.0452   |
| Ordinal-10 sum convolution | 256      | 0.3449   |
| Signed grouped convolution | 32       | 0.00134  |
| Signed grouped convolution | 256      | 0.00416  |

ResNet-18 backbones. The pretrained one-vs-rest count settings use ImageNet-pretrained ResNet-18 with random crop and horizontal flip augmentation, SGD with momentum 0.9, weight decay  $10^{-4}$ , learning rate  $5 \times 10^{-4}$ , and a cosine learning-rate schedule for 500 epochs. The pretrained CIFAR-10 analysis is limited to the evaluated  $n = 16$  and  $n = 64$  one-vs-rest count settings.

**Compute resources.** Reported GPU experiments were run on a single NVIDIA RTX 4090 GPU with 24 GB memory. The CPU dynamic-programming benchmark was run on the paired host CPU for that machine; the exact CPU model was not recorded in the experiment artifact. MNIST-family LeNet runs take approximately 0.4–1.5 GPU-hours per configuration, with peak memory ranging from about 0.4 GB for small-bag settings to 6.5 GB for the largest  $n = 50$  settings. CIFAR-10 pretrained ResNet-18 runs take approximately 8–12 GPU-hours per configuration under the 500-epoch cosine schedule. The reported experiments and backend benchmarks use approximately 150 total GPU-hours; smoke tests and preliminary runs were used for development but are excluded from the reported tables.

## B Mathematical origin: signed counting in $(q, t)$ -Littlewood–Richardson coefficients

The signed-count instance of our framework was originally motivated by a problem in algebraic combinatorics: a conjectural factorization of the  $(q, t)$ -Littlewood–Richardson coefficients arising in the theory of Macdonald polynomials [18]. This object is central across several areas of mathematics—symmetric function theory, representation theory, Schubert calculus on Grassmannians, and partition combinatorics—and it also surfaces in mathematical physics through connections to the Calogero–Sutherland model and related integrable systems [13, 12, 27]. We sketch how this problem reduces to signed counting in our framework; complete mathematical exposition and empirical study are left for future work.

**Setup.** A partition  $\lambda = (\lambda_1, \dots, \lambda_k)$  is identified with its Young diagram, a finite collection of unit cells. For a cell  $u$  in  $\lambda$ , the *arm length*  $a_\lambda(u)$  is the number of cells strictly to the right of  $u$  in the same row, and the *leg length*  $\ell_\lambda(u)$  is the number of cells strictly below  $u$  in the same column. Figure 2 illustrates these quantities on the partition  $(4, 3, 1)$ . The atomic building block of the theory is the rational function

$$b(a, \ell) := \frac{1 - q^a t^{\ell+1}}{1 - q^{a+1} t^\ell}, \quad b_\lambda(u; q, t) := b(a_\lambda(u), \ell_\lambda(u)). \quad (16)$$

**The conjectural factorization.** The Littlewood–Richardson coefficient  $c_{\lambda, \mu}^\nu \in \mathbb{Z}_{\geq 0}$  [13] admits a  $(q, t)$ -refinement  $f_{\lambda, \mu}^\nu(q, t)$ , a rational function in  $q, t$  with  $f_{\lambda, \mu}^\nu(0, 0) = c_{\lambda, \mu}^\nu$ . A long-standing conjecture of Stanley [24] predicts that whenever  $c_{\lambda, \mu}^\nu = 1$ , there exist subsets  $X_{\lambda, \mu}^\nu \subseteq \lambda$ ,  $Y_{\lambda, \mu}^\nu \subseteq \mu$ , and  $Z_{\lambda, \mu}^\nu \subseteq \nu$  such that

$$f_{\lambda, \mu}^\nu(q, t) = \frac{\prod_{u \in Z_{\lambda, \mu}^\nu} b_\nu(u; q, t)}{\prod_{u \in X_{\lambda, \mu}^\nu} b_\lambda(u; q, t) \cdot \prod_{u \in Y_{\lambda, \mu}^\nu} b_\mu(u; q, t)}. \quad (17)$$

The conjecture remains open in general; the case where  $\mu$  is a single-row partition is known explicitly as the *Pieri rule* [18, p. 340].

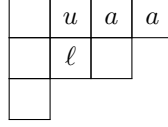


Figure 2: Young diagram of the partition  $(4, 3, 1)$ . For the cell  $u = (1, 2)$ , the arm length  $a_\lambda(u) = 2$  counts the cells marked  $a$  to the right of  $u$  in the same row, and the leg length  $\ell_\lambda(u) = 1$  counts the cell marked  $\ell$  below  $u$  in the same column.

**Reduction to signed finite-support aggregate likelihoods.** Two observations connect Eq. (17) to signed aggregate counting. First, the atomic factor  $b_\lambda(u; q, t)$  depends on the cell  $u$  only through its arm-leg pair  $(a_\lambda(u), \ell_\lambda(u))$ , so cells from different partitions or different positions but sharing the same value  $(a, \ell)$  all contribute the same factor  $b(a, \ell)$  to  $f_{\lambda, \mu}^\nu(q, t)$ , and may therefore be grouped together. Second, the conjectural subsets enter Eq. (17) with opposite multiplicative roles: cells in  $Z_{\lambda, \mu}^\nu \subseteq \nu$  appear in the numerator, while cells in  $X_{\lambda, \mu}^\nu \subseteq \lambda$  and  $Y_{\lambda, \mu}^\nu \subseteq \mu$  appear in the denominator. Letting  $+1$  (resp.  $-1$ ) signify a numerator (resp. denominator) contribution and grouping cells by arm-leg type, the multiplicities collapse into a single signed integer exponent per type, yielding

$$f_{\lambda, \mu}^\nu(q, t) = \prod_{(a, \ell) \in \mathbb{Z}_{\geq 0}^2} b(a, \ell)^{\beta_{\lambda, \mu, \nu, a, \ell}}, \quad \beta_{\lambda, \mu, \nu, a, \ell} = \sum_{u \in \mathcal{B}_{\lambda, \mu, \nu, a, \ell}} s_u z_u. \quad (18)$$

Here  $\mathcal{B}_{\lambda, \mu, \nu, a, \ell} = \{u \in \lambda \sqcup \mu \sqcup \nu : (a_u, \ell_u) = (a, \ell)\}$  is the arm-leg bag, the sign is  $s_u = +1$  for  $u \in \nu$  and  $s_u = -1$  for  $u \in \lambda \sqcup \mu$ , and  $z_u \in \{0, 1\}$  indicates whether the cell  $u$  belongs to the conjectural selected subset  $X_{\lambda, \mu}^\nu \cup Y_{\lambda, \mu}^\nu \cup Z_{\lambda, \mu}^\nu$ . The exponents  $\beta_{\lambda, \mu, \nu, a, \ell}$  are read off from the rational expression of  $f_{\lambda, \mu}^\nu(q, t)$ , while the underlying selections  $\{z_u\}$  are unobserved and need not be unique: distinct latent selections can produce the same vector of observed exponents, exactly the cancellation phenomenon that distinguishes signed counting from plain Bernoulli counting.

Identifying instances with cells, bags with arm-leg equivalence classes, the aggregate label with the observed exponent  $\beta_{\lambda, \mu, \nu, a, \ell}$ , and the latent selection with membership in the conjectural subset, Eq. (18) is exactly the signed finite-support aggregate-likelihood setting of Section 3. Each triple  $(\lambda, \mu, \nu)$  produces a family of such bags indexed by  $(a, \ell)$ , each with a signed integer label, and the convolutional bag-prediction layer of the main paper applies without modification.

**Empirical note.** This signed-count view suggests a future experiment: learning latent cell selections from aggregate exponents under the conjectural factorization. The detailed mathematical formulation, problem-specific architectural choices, and combinatorial diagnostics are outside the scope of the present paper and are left to a separate report.

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